Water Well Condition Prediction in Tanzania

- ML for infrastructure Monitoring
- Predict well status
- Support NGO’S & Government decisions

This project focuses on predicting the condition of water wells in Tanzania using machine learning.
The goal is to help organizations identify wells that are non-functional or at risk, so they can prioritize repairs and improve access to clean water.

Business Understanding

- Tanzania faces challenges in maintaining water infrastructure. Many wells fail over time, and resources are limited.
  This project aims to help decision-makers identify which wells need attention most urgently:
  - Limited access to clean water
  - Many wells are non functional or deteriorating
  - Need for efficient resource allocation

Data Understanding

- The dataset contains detailed information about wells, including location, construction, and operational factors.
  The target variable classifies wells into three categories based on functionality.
- 59,400 water wells
- Features: location, pump&water charactetristics ,construction details etc
- Predictive variable: Functional, needs repair or non functional pumps.

Algorithm Modelling

- I built a full machine learning pipeline to ensure consistent preprocessing.
  Multiple models were tested to identify the best-performing approach.
- Data preprocessing pipeline: missing value imputation, feature engineering, encoding categorical variables
- Models used: Random Forest, Extra Trees & Gradient Boosting

Model Evaluation

- The Random Forest model performed best, achieving about 80% accuracy.
  However, the model struggles more with identifying wells that need repair, likely due to fewer examples in the dataset.
- Random Forest is the best model
- Accuracy: 80.4%
- Strong performance: functional wells & non functional wells
- Challenge: lower accuracy for “needs repair”

Key Insights

- The model revealed that geography plays a major role in well performance.
  Additionally, older wells and those with low water quantity are more likely to fail.
- Location is the strongest predictor
- Older wells fail more often
- Water quantity strongly affects functionality
- High population increases strain

Recommendations

- Based on the model, resources should be focused on regions with higher failure rates.
  Preventative maintenance on older wells can also reduce long-term costs.
- Prioritise high risk regions
- Repair aging infrastructure
- Monitor wells with low water levels
- Focus on high demand areas

Project’s Next Steps

- Future improvements could focus on better identifying wells that are close to failure.
  This model could also be deployed as a real-time decision-support system.
- Improve detection of ‘needs repair’ wells
- Perform geospatial analysis
- Tune models further
- Deploy as a decision tool

Thankyou.

Project by: Mercy Apondi